Plug and Play Multiseat

Linux Plumbers Conference 2009
BoF Session
Bernie Thompson

About the speaker

- "Linux vs. NT and OS/2" (Issue 1 LJ, 1994)
- S3's graphics driver guy on-site at Msft
- Cosource & Applix Linux Division
- Windows Development Manager @ Msft
- DisplayLink and now plugable

- Clearly a checkered past!

Multiseat

- Awareness
- What defines multiseat
  - “Terminals” are just collections of devices
  - Sharing computing power of central PC
- Size of multiseat deployments
  - Linux: hundreds of thousands
  - Windows: over a million (nComputing, etc.)

4 computers for the family







- Dr. Quentin Stafford-Fraser and others
- Vision for low-cost computing for the last few billion people on the planet
- Good background at http://ndyio.org/
- Founded DisplayLink as commercial arm
- Q one of the original developers of VNC

Internet Cafe scenario



ndiyo!

South Africa



ndiyo!

No more fighting over PCs



Dirt cheap

- $299 Atom 1.6Ghz netbook (2GB RAM)
- Driving 4 USB terminals
- Good enough for web browsing and most of Edubuntu collection of apps

- Wormux a favorite app – explosions, joy!
- Also good for Open Office and Plumbers Conference presentations

Real Experiences at Home

- With kids, with Edubuntu applications

- Windows → Linux easy from UI perspective
- Tough from app compat perspective
- USB seats: Still some things to improve

- Deploy at school? A lot to think about.

USB Multiseat

Traditional

- PCIe Graphics
- Higher Gfx Perf
- Similar hw cost
- **Must open box**
- VGA/Display cable
- Limited – free slots

USB

- All devices on USB
- Limited until USB3
- +$20 to USB dock
- **Plug and play**
- WUSB, CAT5
- Go crazy!

Demo

(Murphy begone)

Zero-state, No CPU, All USB





5V 2.5A (~7 port USB hub)



USB Graphics

- DisplayLink (displaylink_drv.so - GPL)
- MCT (sisusb_drv.so - GPL)
- Both have drivers for the adventurous
- More vendors coming, esp with USB 3.0

- Different from traditional 2D/3D graphics controller. Think remote framebuffer.

Still early days

- Many todos and outstanding problems
- Common issues across all multiseat
- Unique issues/opportunties for udev integration – taking advantage of USB plug and play, getting power management right
- Integration with new ConsoleKit

Code/script walkthrough

- This is a kernel developer's conference
- Only I'm probably least qualified in room …

- http://libdlo.freedesktop.org/wiki/MultiSeatTerminal
  - End-user instructions
  - And some helpful scripts

- Notes on opportunities to improve very welcome!

Q&A

plugable



- A “next generation device company”
- $99 zero-state dock/thin client
  - http://www.amazon.com/gp/product/B002PONXAI
- USB2 multifunction device with graphics, audio, eth, etc.
- Every chip/function on it has some level of open source driver – enabling interesting new uses for the device
- http://getsatisfaction.com/ support
- More products coming with this kind of philosophy

BoFing (time permitting)

- Break into four groups
  - udev/kernel ideas
  - GDM/X ideas
  - Distro integration ideas
  - Interesting use case ideas
- Group leader reports back with top 3